Logistic Regression with Regularisation

CF

$$J(\theta_0, \theta_1) = -y \log(h_\theta(x_i)) - (1-y_i)\log(1-h_\theta(x_i))$$

$$h_\theta(x) = \frac{1}{1+e^{-(\theta_0+\theta_1 x_1)}}$$

Ridge

$$J(\theta_0, \theta_1) = -y \log(h_\theta(x_i)) - (1-y_i)\log(1-h_\theta(x_i)) + \underbrace{L_2 \text{ regularisation}}_{\Downarrow \text{ reduce overfitting}}$$

lasso

$$J(\theta_0, \theta_1) = -y \log(h_\theta(x_i)) - (1-y_i)\log(1-h_\theta(x_i)) + \underbrace{L_1 \text{ regularisation}}_{\Downarrow \text{ feature selection}}$$

Elastic net

$$J(\theta_0, \theta_1) = -y \log(h_\theta(x_i)) - (1-y_i)\log(1-h_\theta(x_i)) + \underbrace{L_1 \text{ regularisation}}_{\Downarrow \text{ reduce overfitting}} + \underbrace{L_2 \text{ regularisation}}_{\Downarrow \text{ feature selection}}$$

L_2 regularisation $\Rightarrow$ reduces Overfitting

$$J(\theta_0,\theta_1) = -y_i \log h_\theta(x)) - (1-y)\log(1-h_\theta(x) + \lambda \sum_{i=1}^{n} (\text{slope})^2$$

L_1 regularization $\Rightarrow$ feature selection

$$J(\theta,\theta_1) = -y \log(h_\theta(x)) - (1-y)\log(1-h_\theta(x)) + \lambda \sum_{i=1}^{n} |\text{slope}|$$

Elastic Net (L_1+L_2)

$$J(\theta_0,\theta_1) = -y_i \log(h_\theta(x)) - (1-y_i)\log(1-h_\theta(x)) + \underbrace{\lambda_1 \sum_{i=1}^{n} (\text{slope})^2}_{L_2} + \underbrace{\lambda_2 \sum_{i=1}^{n} |\text{slope}|}_{L_1}$$

$$\boxed{C = \frac{1}{\lambda}}$$

$\rightarrow$ regularisation parameter in logistic regression Scikit implementation

(Earlier — alpha)